

IMAMUL AHSAN

Research Software Engineer | Empirical Software Engineering | Human-Centered Systems

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RESEARCH PROFILE

Research-oriented software engineer with experience in empirical user studies, interactive systems, industrial software engineering, and data-driven software platforms. Currently completing a Master's thesis on configurable multimodal feedback systems for human-robot collaboration involving usability evaluation, interaction analysis, and empirical experimentation. Interested in PhD research in empirical software engineering, usable software systems, human-centered systems, and interactive software platforms.

RESEARCH INTERESTS

Empirical Software Engineering; Human-Computer Interaction (HCI); Human-Robot Interaction (HRI); Interactive Systems; User Studies; Human-Centered Computing.

EDUCATION

Technische Universität Chemnitz

M.Sc. Automotive Software Engineering
Current grade: 2.47 before thesis evaluation.

Chemnitz, Germany
Apr 2021 – Expected Jun 2026

Islamic University of Technology (IUT)

B.Sc. Computer Science and Engineering

Bangladesh
Nov 2014 – Nov 2018

MASTER THESIS

Development of a Web Interface for a Human-Robot Workplace

Technische Universität Chemnitz

Expected submission: Jun 2026

- Designed and developed a configurable web-based interface for multimodal feedback in collaborative robotic assembly environments.
- Conducted a two-phase empirical user study investigating transparency, usability, cognitive workload, and user-driven configuration behavior.
- Collected and analyzed questionnaire responses, interaction logs, and user behavior using descriptive statistics and Wilcoxon signed-rank tests.
- Evaluated configurable software behavior and user interaction patterns in collaborative software systems through empirical analysis.
- Integrated web-based frontend systems with ROS-based robotic infrastructure through REST APIs and runtime reconfiguration mechanisms.

Methods and technologies: Empirical user studies, usability evaluation, hypothesis testing, descriptive statistics (Python, pandas, scipy), Angular, FastAPI, REST APIs, ROS integration.

PUBLICATIONS

- Kushol, R., **Ahsan, I.**, & Raihan, M. N. (2018). "An Android-Based Useful Text Extraction Framework Using Image and Natural Language Processing." *International Journal of Computer Theory and Engineering*, 10(3), 77–83.
- Sobhan, M., Chowdhury, M. Z., **Ahsan, I.**, Mahmud, H., & Hasan, M. K. (2019). "A Communication Aid System for Deaf and Mute using Vibrotactile and Visual Feedback." In *2019 International Seminar on Application for Technology of Information and Communication (iSemantic)*, IEEE, pp. 184–190.

PROFESSIONAL EXPERIENCE

Bosch Rexroth

Research Software Engineer / Working Student

Ulm, Germany
Mar 2024 – Present

- Develop and maintain industrial software applications supporting internal business processes and configurable software workflows.
- Design RESTful APIs and backend services for enterprise systems using MySQL-based data integration.
- Contributed to development of configurable and user-oriented software systems supporting industrial workflows and cross-functional collaboration.
- Improve backend performance and database efficiency, contributing to approximately 20% performance optimization.
- Contribute to CI/CD pipelines and containerized deployment workflows using GitLab CI/CD and Docker.
- Collaborate in agile cross-functional teams involving software engineering, business stakeholders, and system integration tasks.

Prime University

Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh
Feb 2019 – May 2021

- Taught undergraduate courses including Software Engineering, Artificial Intelligence, Web Development, and Database Systems.
- Supervised student software projects involving user-facing applications and applied machine learning systems.
- Supported academic mentoring, student evaluation, and course planning activities.

SELECTED PROJECTS

AI Language Tool using OpenAI API

- Developed an AI-assisted language platform supporting summarization, grammar checking, translation, and educational content generation.

Chemnitz Interactive Map

- Built a location-based interactive platform integrating OpenStreetMap services and user-managed points of interest.

TECHNICAL SKILLS

Research Methods	Empirical studies, usability evaluation, interaction analysis, statistical analysis
Programming	Python, JavaScript, TypeScript, Java, PHP
Frameworks	Angular, Node.js, Express, FastAPI
Software Systems	REST APIs, configurable software systems, backend systems, software integration
Databases	MySQL, PostgreSQL, MongoDB
DevOps & Infrastructure	Docker, GitLab CI/CD, Git, Linux, Jira
Areas	Empirical software engineering, software sustainability, human-centered systems, interactive systems
Data Analysis	pandas, scipy, descriptive statistics, hypothesis testing

CERTIFICATIONS

- DevOps on AWS Specialization
- Meta Front-End Developer Certificate

LANGUAGES

English: C1 | German: A2/B1 (in progress) | Bengali: Native